



SECURE • DETECT • REVIEW

AI-Powered Vulnerability Detection & Secure Code Review Platform

An AI system that scans source code, classifies vulnerabilities by severity, and generates secure, corrected code with a downloadable review report.

PROJECT REVIEW PRESENTATION

What the platform is built to achieve

- 01** Automatically scan uploaded source code (Java, Python, C++, and more) using AI
- 02** Detect and classify vulnerabilities with clear severity levels
- 03** Explain the real-world risk behind every flagged issue
- 04** Recommend secure coding practices and AI-corrected code snippets
- 05** Generate a downloadable PDF security review report

System architecture overview

Presentation Layer

Web UI — code upload, dashboards, report viewer/download

Application / API Layer

Handles requests, file parsing, orchestrates scan jobs

AI / ML Analysis Engine

Static analysis + LLM-based reasoning to detect & explain vulnerabilities

Data Layer

Stores scan history, vulnerability rules, and generated reports

Code Upload & Analysis

Multi-language upload

Accepts source files in Java, Python, C++, and other common languages.

Automated AI scan

The AI engine scans the uploaded code the moment it arrives — no manual trigger needed.

Error + vulnerability detection

Flags both coding errors and potential security vulnerabilities in a single pass.

Instant issue display

Detected issues are displayed clearly, ready for classification and review.

Vulnerability Detection & Risk Assessment

 LOW

 MEDIUM

 HIGH

 CRITICAL

Severity is assigned to every detected vulnerability, so the most dangerous issues get attention first.

Classifies Vulnerabilities

SQL Injection, XSS, Buffer Overflow, Hardcoded Passwords, and more.

Explains the Risk

Plain-language reasoning on why each vulnerability is dangerous.

Security Risk Report

Consolidates every finding into a structured risk report.

Secure Code Review & Recommendations

- Suggests secure coding practices for each finding
- Provides AI-generated corrected code snippets
- Generates a final secure code review report
- Lets users download the report as a PDF

BEFORE → AFTER

VULNERABLE

```
query = "SELECT * FROM users WHERE id=" + userId
```

SECURE FIX

```
query = "SELECT * FROM users WHERE id=?"  
cursor.execute(query, [userId])
```